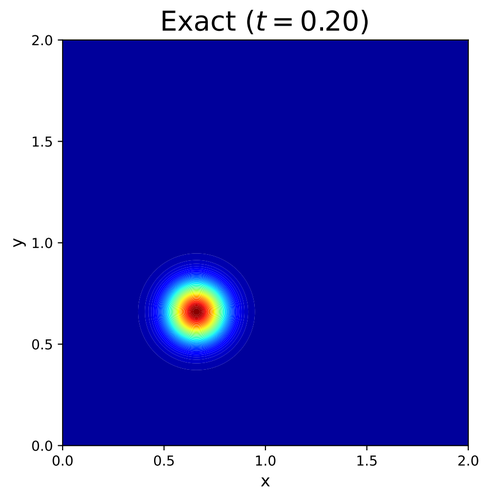
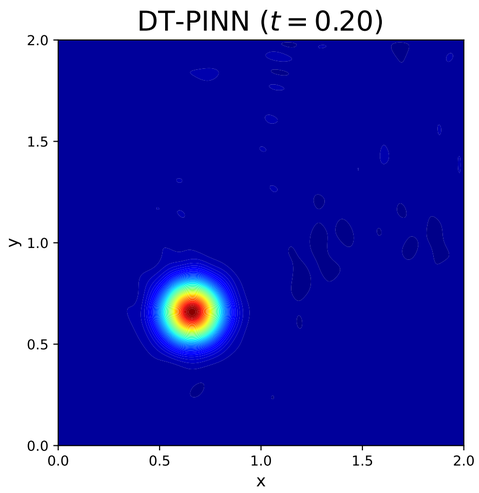
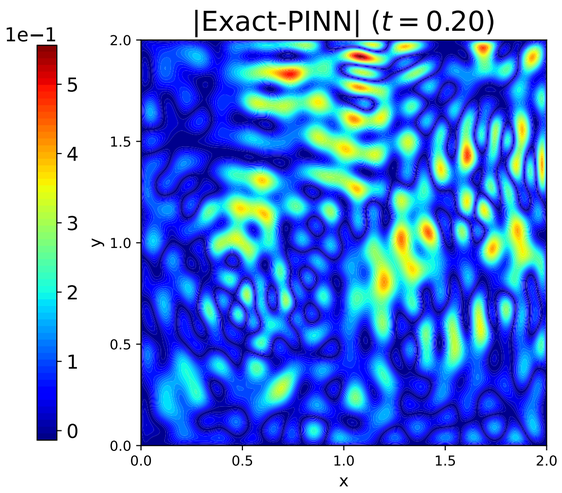
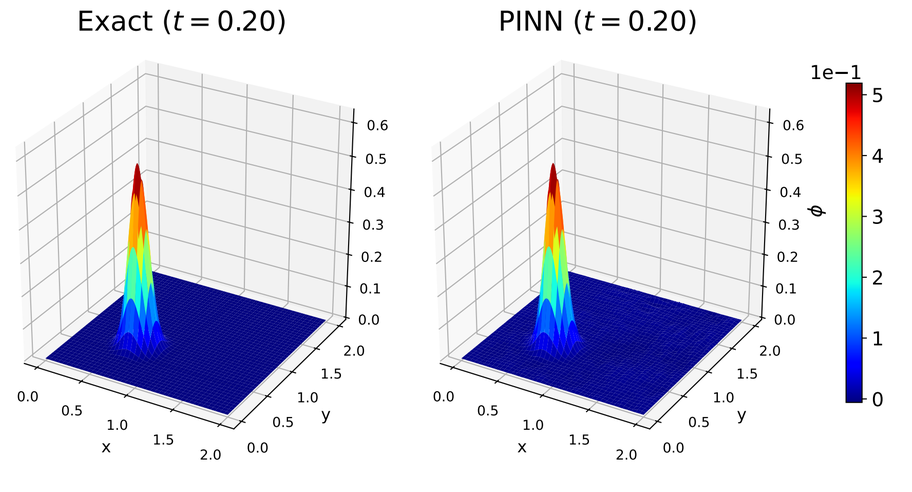

Exact ($t = 0.20$)
x
y


DT-PINN ($t = 0.20$)
x
y


|Exact-PINN| ($t = 0.20$)
1e−1
1e−3
x
y

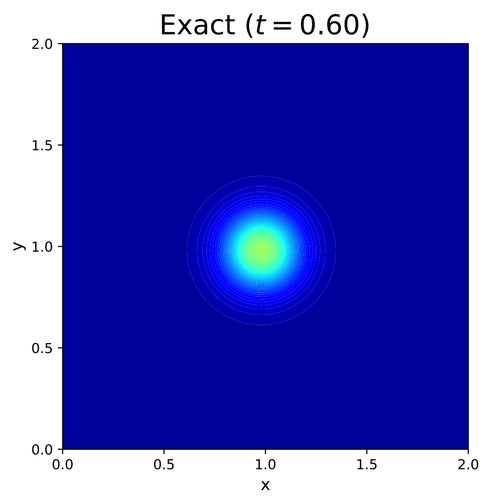

Exact ($t = 0.20$)
PINN ($t = 0.20$)
1e−1
x
y
ϕ

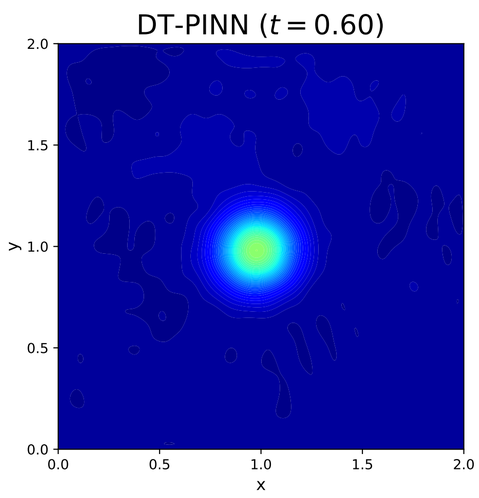
Exact ($t = 0.40$)
x
y

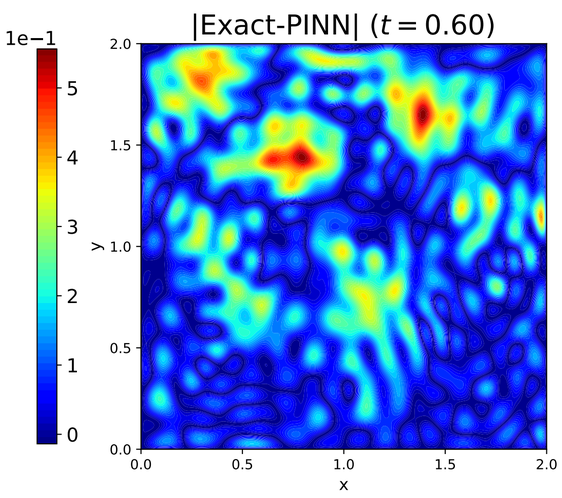
DT-PINN ($t = 0.40$)
x
y

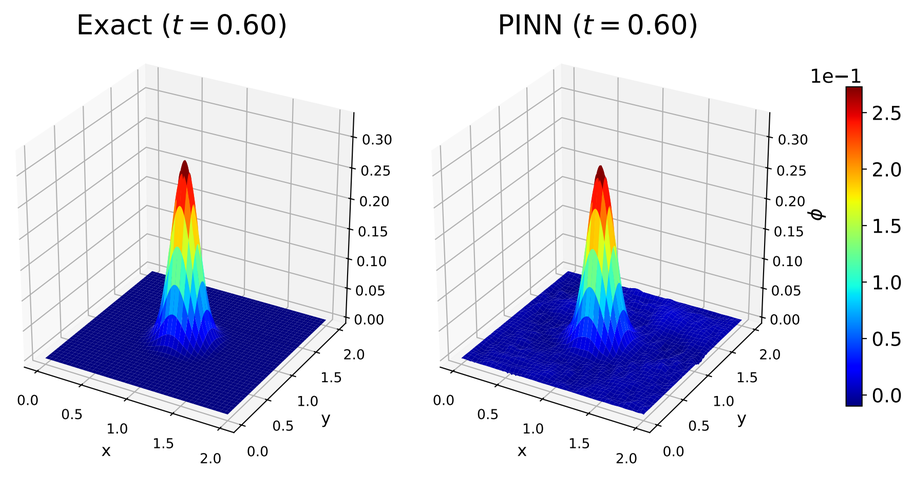
|Exact-PINN| ($t = 0.40$)
1e−1
1e−2
x
y

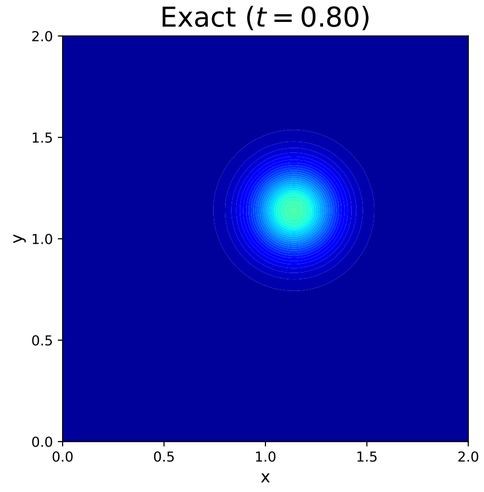
Exact ($t = 0.40$)
PINN ($t = 0.40$)
1e−1
x
y
ϕ

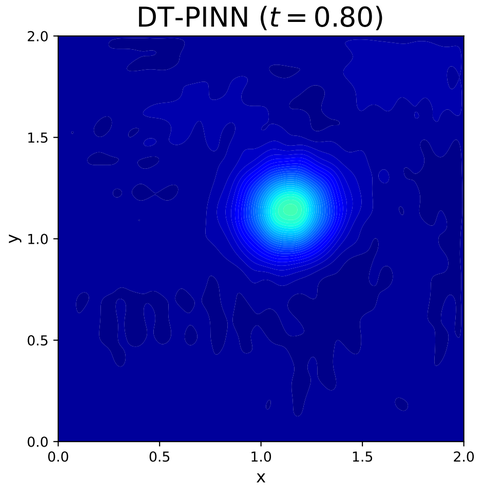

Exact ($t = 0.60$)
x
y

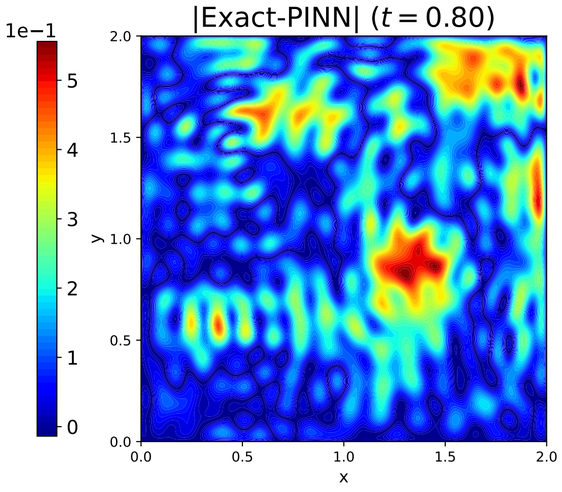

DT-PINN ($t = 0.60$)
x
y

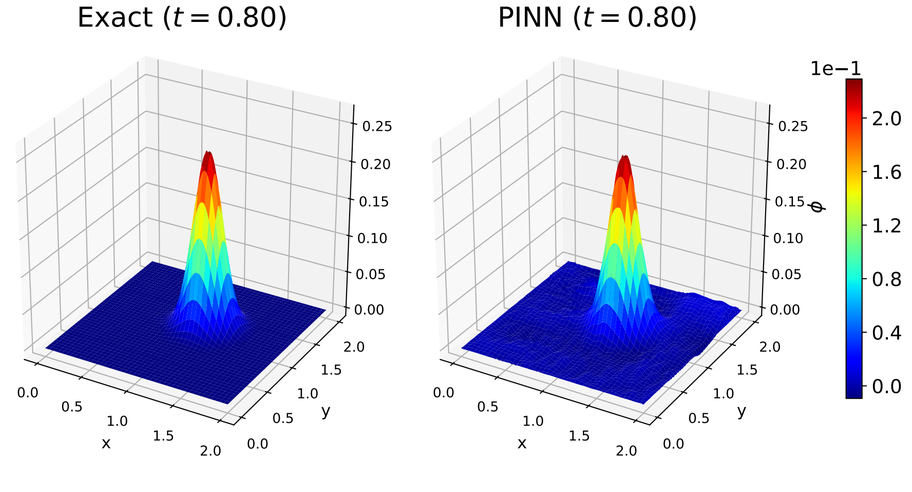

|Exact-PINN| ($t = 0.60$)
1e−1
1e−2
x
y

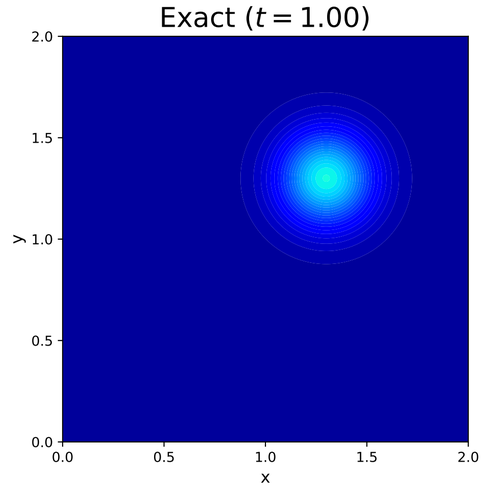

Exact ($t = 0.60$)
PINN ($t = 0.60$)
1e−1
x
y
ϕ

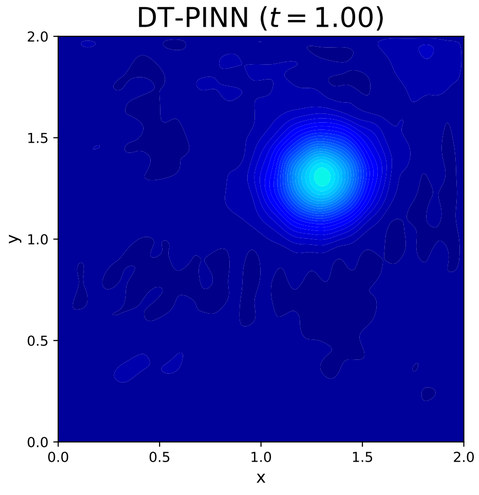

Exact ($t = 0.80$)
x
y

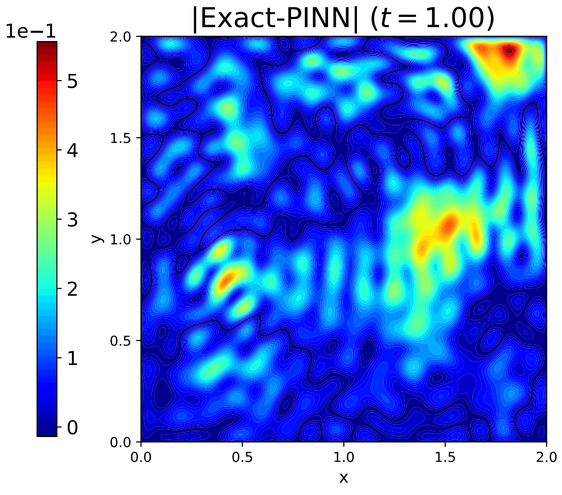

DT-PINN ($t = 0.80$)
x
y

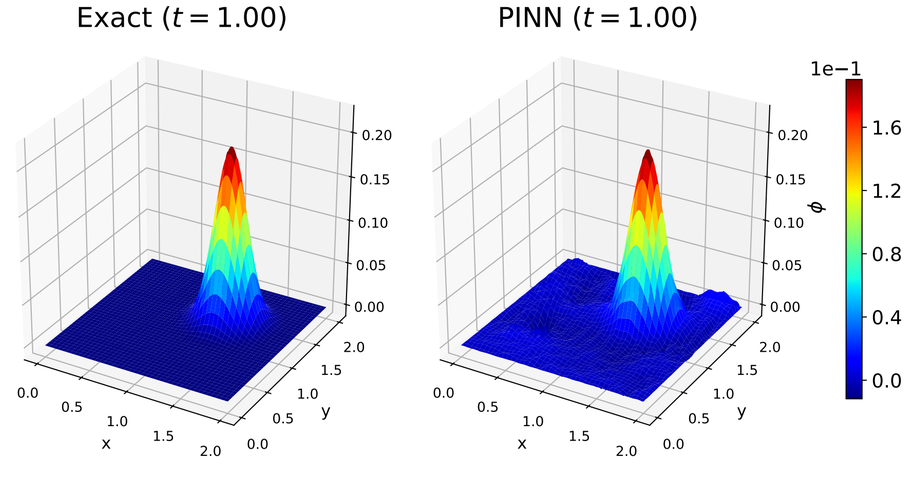

|Exact-PINN| ($t = 0.80$)
1e−1
1e−2
x
y


Exact ($t = 0.80$)
PINN ($t = 0.80$)
1e−1
x
y
ϕ


Exact ($t = 1.00$)
x
y


DT-PINN ($t = 1.00$)
x
y


|Exact-PINN| ($t = 1.00$)
1e−1
1e−2
x
y


Exact ($t = 1.00$)
PINN ($t = 1.00$)
1e−1
x
y
ϕ